

Generative Modelling (MSc)

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Outline

Graphical models

- Random variables and probabilities
- Graphical model
- Conditional independence

Example Machine Learning Tasks

- Classification: Discriminative modelling
- Classification: Generative modelling
- Density estimation

Algorithms for latent variable models

- Gradient algorithms
- Expectation maximisation
- Bayesian Inference

Exercises

- Density estimation
- Classification

Graphical models

- Random variables and probabilities

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- Conditional independence

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- Classification

Notational problems

In probability theory, we are dealing with functions P on sets, called measures. In statistics, we are dealing with random variables. While these two can be related, it is common to use shorthand notation in statistics. If in doubt, use the following conversion table.

Probability theory

1. Sets $A \subset \Omega$ are **events**.
2. Probability measure P on Ω .
3. $P(A) \in [0, 1]$ with $A \subset \Omega$
4. $P(A \cap C)$ the probability $\omega \in A \cap C$
5. $P(A|B)$ the probability of A if B is true.
6. Marginal distribution

$$P(A) = \sum_{i=1}^n \mathbb{P}(A \cap H_i)$$
7. Bayes's theorem:

$$P(A | B) = P(A \cap B) / P(B).$$

Statistics

1. Random variables $x : \Omega \rightarrow X$.
2. The measure P is **implicit**.
3. $\mathbb{P}(x)$ for the distribution of x
4. $\mathbb{P}(x, y)$ the joint distribution of x, y .
5. $\mathbb{P}(x|y)$ the distribution of x if y is given.
6. Marginal distribution

$$\mathbb{P}(y) = \sum_{i \in x} \mathbb{P}(x = i, y).$$
7. Bayes's theorem:

$$\mathbb{P}(x|y) = \mathbb{P}(x, y) / \mathbb{P}(y)$$

Probability and statistics

Probability theory

Here, we talk a lot about **probability measures** P on some space Ω . These are functions on **sets**, so that

- ▶ $P(\Omega) = 1$
- ▶ $P(A) \in [0, 1]$ for any $A \subset \Omega$
- ▶ $P(A \cup B) = P(A) + P(B)$ if $A \cap B = \emptyset$.

Statistics

In statistics, we are interested in random variables $x, y, \dots$ which are **functions** $x : \Omega \rightarrow X$, $y : \Omega \rightarrow Y$ etc.

- ▶ We write $\mathbb{P}(x)$ as a shorthand for *the probability distribution of x* .
- ▶ The distribution of x arises in the following way:
 1. Sample ω from P
 2. Calculate $x(\omega)$.
- ▶ We can get a **probability measure** P_x by measuring how often x falls in various sets $A \subset X$:

$$P_x(A) \triangleq P(\{\omega : x(\omega) \in A\})$$

Here, $\{\omega : x(\omega) \in A\}$ is literally the set of random outcomes ω for which $x(\omega)$ is in the set A . To avoid writing this cumbersome expression, we simply write $\mathbb{P}(x)$ as a general shorthand. This allows us to also write

$$\mathbb{P}(x \in A) = P_x(A).$$

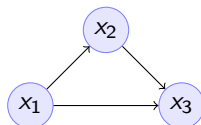
Joint distributions, multiple variables

If you thought things were messy, wait to see what happens when you have more than one variable!

- ▶ $x : \omega \rightarrow X, y : \omega \rightarrow Y$.
- ▶ The measure $P_{x,y}$ has to be defined on the subsets of $X \times Y$.

$$P_{x,y}(A) = P(\{\omega : (x(\omega), y(\omega)) \in A\})$$

Graphical models



Directed acyclic graph

A directed acyclic graph with

- ▶ Nodes $x_1, \dots, x_n$
- ▶ Edges $x_i \rightarrow x_j$.

so that there are no cycles in the graph.

Conditional independence from graphical models

- ▶ Each **node** of the model corresponds to a **random variable**
- ▶ The **parent** of a node are the **direct dependencies** of the random variable.

Model specification



- ▶ The graphical model tells us **what depends on what**.
- ▶ It does not tell us **how to generate data**.
- ▶ We need to specify the **functional relationship** between variables.

Statistics

$$\begin{aligned}
 x &\sim f \\
 y \mid x = a &\sim g(a) \\
 z \mid y = b &\sim h(b)
 \end{aligned}$$

Python

```

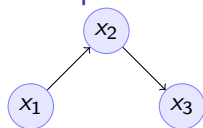
x = f()
y = g(x)
z = h(y)

```

Graphical models and factorisation

- ▶ Graphical models tell us what directly depends on what
- ▶ They allow us to simplify the **joint distribution** of the variables
- ▶ This is formed into a **product of factors**, one for each variable.

Example



This graphical model implies the factorisation

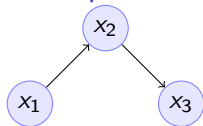
$$\mathbb{P}(x_1, x_2, x_3) = \mathbb{P}(x_3 \mid x_2, x_1) \mathbb{P}(x_2 \mid x_1) \mathbb{P}(x_1) = \mathbb{P}(x_3 \mid x_2) \mathbb{P}(x_2 \mid x_1) \mathbb{P}(x_1)$$

Notice that for each factor $\mathbb{P}(x_i \mid x_j)$, x_j is the **parent** of x_i .

Conditional independence

- ▶ Graphical models tell us what **directly** depends on what
- ▶ Consequently, they also specify **conditional independence**

Example



$$\mathbb{P}(x_1, x_2, x_3) = \mathbb{P}(x_3 \mid x_2) \mathbb{P}(x_2 \mid x_1) \mathbb{P}(x_1)$$

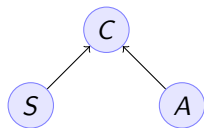
Definition (Conditional independence)

We say that variables x, y are **conditionally** independent given z and write $x \perp\!\!\!\perp y \mid z$ if and only if

$$\mathbb{P}(x, y \mid z) = \mathbb{P}(x \mid z) \mathbb{P}(y \mid z)$$

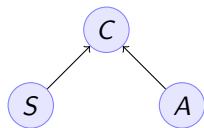
- ▶ In the above example, it holds that $x_3 \perp\!\!\!\perp x_1 \mid x_2$.

Smoking and lung cancer



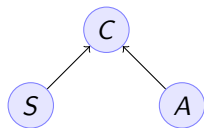
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- ▶ S: Smoking, C: cancer, A: asbestos exposure.

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Smoking and lung cancer

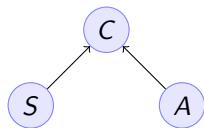


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XOR example

- ▶ $C = \text{xor}(S, A)$.

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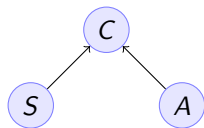


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- ▶ If we know $C = 1$, $S = 1$, what is A ?

Smoking and lung cancer

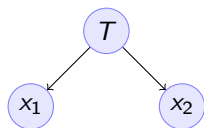


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- ▶ $C = \text{xor}(S, A)$.
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- ▶ C explains away

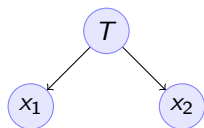
Time of arrival at work



Time of arrival at work graphical model where T is a traffic jam and x_1 is the time John arrives at the office and x_2 is the time Jane arrives at the office.

- ▶ Even though x_1, x_2 are **not independent**, they become independent once you know T , i.e. $x_1 \perp\!\!\!\perp x_2 \mid T$.

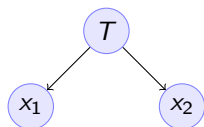
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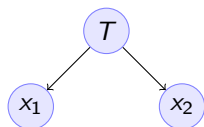
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- ▶ $\mathbb{P}(x_1, x_2, T) = \mathbb{P}(x_2 \mid T) \mathbb{P}(x_1 \mid T) \mathbb{P}(T)$ from the graph.

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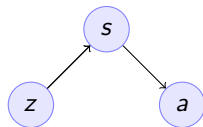
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- ▶ $\mathbb{P}(x_1, x_2, T) = \mathbb{P}(x_2 \mid T) \mathbb{P}(x_1 \mid T) \mathbb{P}(T)$ from the graph.
- ▶ $\mathbb{P}(x_1, x_2, T) / \mathbb{P}(T) = \mathbb{P}(x_1, x_2 \mid T) = \mathbb{P}(x_2 \mid T) \mathbb{P}(x_1 \mid T)$. □

School admission

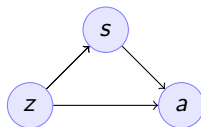
Example

School	Male	Female
A	62	82
B	63	68
C	37	34
D	33	35
E	28	24
F	6	7
Average	50	27

- ▶ z : gender
- ▶ s : school applied to
- ▶ a : admission



Is admission
independent of gender?



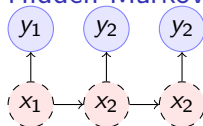
How about here?

Time-series

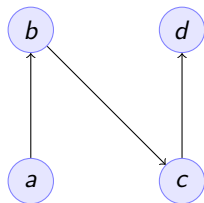
Markov chain



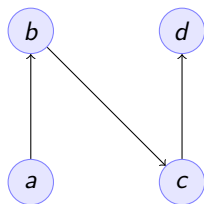
Hidden Markov model



What is the model for this graph?

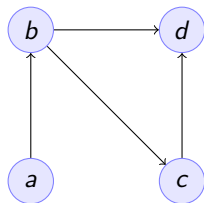


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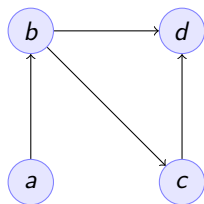


$$\mathbb{P}(a, b, c, d) = \mathbb{P}(d|c) \mathbb{P}(c|b) \mathbb{P}(b|a) \mathbb{P}(a)$$

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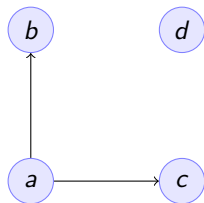


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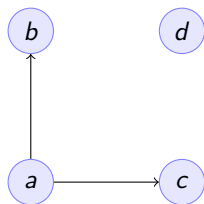


$$\mathbb{P}(a, b, c, d) = \mathbb{P}(d|b, c) \mathbb{P}(c|b) \mathbb{P}(b|a) \mathbb{P}(a)$$

What is the model for this graph?

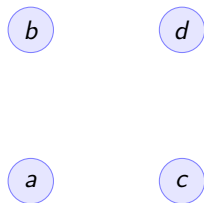


What is the model for this graph?



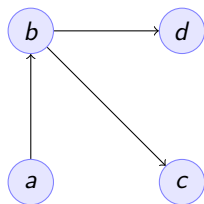
$$\mathbb{P}(a, b, c, d) = \mathbb{P}(d) \mathbb{P}(c|a) \mathbb{P}(b|a) \mathbb{P}(a)$$

Draw the graph for this model



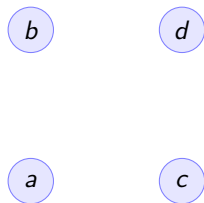
$$P(a, b, c, d) = P(a)P(b|a)P(c|b)P(d|b)$$

Draw the graph for this model



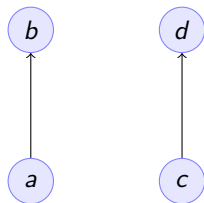
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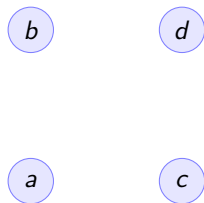
$$P(a, b, c, d) = P(a)P(b|a)P(d|c)P(c)$$

Draw the graph for this model



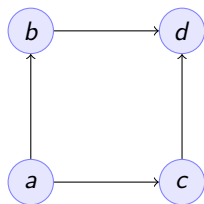
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Conditional independence (general)

- ▶ Consider variables $x_1, \dots, x_n$.
- ▶ Let B, D be subsets of $[n]$, and
- ▶ $x_B \triangleq (x_i)_{i \in B}$ be the variables with indices in B .
- ▶ $x_{-j} \triangleq (x_i)_{i \neq j}$ all the variables apart from x_j .

Definition (Conditional independence)

We say x_i is **conditionally independent** of x_B given x_D and write

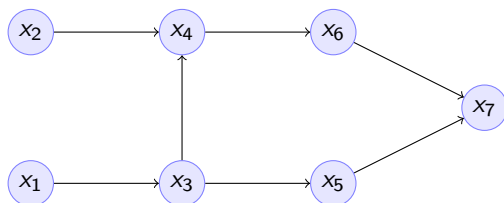
$$x_i \perp\!\!\!\perp x_B \mid x_D$$

if and only if:

$$\mathbb{P}(x_i, x_B \mid x_D) = \mathbb{P}(x_i \mid x_D) \mathbb{P}(x_B \mid x_D).$$

- ▶ For this to hold in graphical model, D must separate i from B in the graph.

More complex example



In this example, we have:

$$x_7 \perp\!\!\!\perp x_1, x_2 \mid x_3, x_4$$

and

$$x_7 \perp\!\!\!\perp x_3 \mid x_4, x_5$$

Graphical models

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Graphical model

Conditional independence

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Generative modelling

General idea

- ▶ Data (x_t, y_t) .
- ▶ Need to model $P(y|x)$.
- ▶ Model the **complete** data distribution: $P(x|y)$, $P(x)$, $P(y)$.
- ▶ Calculate $P(y|x) = \frac{P(x|y)P(y)}{P(x)}$.

Examples

- ▶ **Naive Bayes** classifier.
- ▶ **Gaussian mixture** model.
- ▶ Large language models.

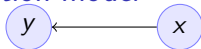
Modelling the data distribution in classification

- ▶ Need to estimate the density $P(x|y)$ for each class y .
- ▶ Need to estimate $P(y)$.

The basic graphical model

A discriminative classification model

Here $P(y|x)$ is given directly.



A generative classification model

Here $P(y|x) = P(x|y)P(y)/P(x)$.



An unsupervised generative model

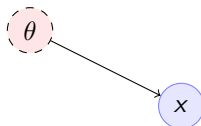
Here we just have $P(x)$.



Adding parameters to the graphical model

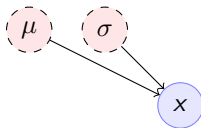
- ▶ We can also see the parameters of the distribution as (random) variables.
- ▶ We can put those random variables in the graphical model as well.
- ▶ Since the parameters are not observed, we denote them with dashed circles.
- ▶ They are a type of **latent** or **hidden** variable.

A Bernoulli RV



Here, $x|\theta \sim \text{Bernoulli}(\theta)$

A normally distributed variable



Here $x|\mu, \sigma \sim \text{Normal}(\mu, \sigma^2)$

Classification: Naive Bayes Classifier

- ▶ Data (x, y)
- ▶ $x \in X$
- ▶ $y \in Y \subset \mathbb{N}$, N_i : amount of data from class i .

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Separately model each class

- ▶ Assume each class data comes from a different normal distribution
- ▶ $x|y = i \sim \text{Normal}(\mu_i, \sigma_i I)$
- ▶ For each class, calculate
 - ▶ Empirical mean $\hat{\mu}_i = \sum_{t: y_t = i} x_t / N_i$
 - ▶ Empirical variance $\hat{\sigma}_i$.

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Decision rule

Use Bayes's theorem:

$$P(y|x) = P(x|y)P(y)/P(x),$$

choosing the y with largest posterior $P(y|x)$.

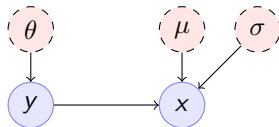
- ▶ $P(x|y = i) \propto \exp(-\|\hat{\mu}_i - x\|^2 / \hat{\sigma}_i^2)$

Graphical model for the Naive Bayes Classifier

When $x \in \mathbb{R}$

Assume k classes, then

- ▶ $\mu = (\mu_1, \dots, \mu_k)$
- ▶ $\sigma = (\sigma_1, \dots, \sigma_k)$
- ▶ $\theta = (\theta_1, \dots, \theta_k)$



- ▶ $y \mid \theta \sim \text{Mult}(\theta)$
- ▶ $x \mid y, \mu, \sigma \sim \text{Normal}(\mu_y, \sigma_y^2)$

Density estimation

The simplest type of generative model is just modelling the distribution of x . There are a number of models for this.

Parametric models

- ▶ Fixed histograms
- ▶ Gaussian Mixtures

Non-parametric models

- ▶ Variable-bin histograms
- ▶ Infinite Gaussian Mixture Model
- ▶ Kernel methods

Histograms

Fixed histogram

- ▶ Hyper-Parameters: number of bins
- ▶ Parameters: Number of points in each bin.

Variable histogram

- ▶ Hyper-parameters: Rule for constructing bins
- ▶ Generally $\sqrt{n}$ points in each bin.

Gaussian Mixture Model

Hyperparameters:

- ▶ Number of Gaussian k .

Parameters:

- ▶ Multinomial distribution θ over Gaussians
- ▶ For each Gaussian i , center μ_i , covariance matrix Σ_i .

Algorithms:

- ▶ Expectation Maximisation
- ▶ Gradient Ascent
- ▶ Variational Bayesian Inference (with appropriate prior)

Details of Gaussian mixture models

Model. For each point x_t :

- ▶ $z_t \mid \theta \sim \text{Mult}(\theta_i)$, $\theta \in \Delta^k$
- ▶ $x_t \mid z_t = i \sim \text{Normal}(\mu_i, \Sigma_i)$.
- ▶ $\text{Mult}(\theta)$ is **multinomial**

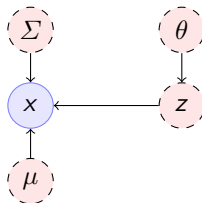
$$\mathbb{P}(z_t = i \mid \theta) = \theta_i$$

- ▶ $\text{Normal}(\mu, \Sigma)$ is **multivariate Gaussian**

$$p(x \mid \mu, \Sigma) \propto \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right)$$

- ▶ The generating distribution is

$$p(x \mid \theta, \mu, \Sigma) = \sum_{z \in [k]} p(x \mid \mu_z, \Sigma_z) P(z \mid \theta).$$



Applications of Gaussian mixture models

- ▶ Density estimation
- ▶ Clustering
- ▶ Used as part of a more complex model.

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Expectation maximisation

Bayesian Inference

Exercises

Density estimation

Classification

Gradient ascent

In the following we use θ for all the parameters of the Gaussian mixture model, with $x = (x_1, \dots, x_T)$ and $z = (z_1, \dots, z_T)$

Objective function

One way to estimate θ is through maximising the likelihood $L(\theta) = P(x|\theta)$

Marginalisation over latent variable

However, we need to marginalise over all values z

$$L(\theta) = \sum_z P(z, x|\theta)$$

For T data points and k different values of z_t , there are k^T vectors z to sum over.

Gradient ascent

If we can calculate the gradient of L , we can use gradient ascent to update our parameters:

$$\theta^{(n+1)} = \theta^{(n)} + \alpha \nabla_{\theta} L(\theta).$$

Gradient calculation

Here we use the **log trick**: $\nabla \ln f(\theta) = \nabla f(\theta)/f(\theta)$.

$$\nabla_{\theta} L(\theta) = \sum_z \nabla_{\theta} P(z, x | \theta) \quad (1)$$

$$= \sum_z P(z, x | \theta) \nabla_{\theta} \ln P(z, x | \theta) \quad (2)$$

$$= \sum_z P(x | z, \theta) P(z | \theta) \nabla_{\theta} \ln P(z, x | \theta) \quad (3)$$

$$\approx \frac{1}{m} \sum_{i=1}^m P(x | z^{(i)}, \theta) \nabla_{\theta} \ln P(z^{(i)}, x | \theta) \quad z^{(i)} \sim P(z | \theta) \quad (4)$$

The final approximates the sum with the sample mean, sampling $z^{(i)}$ from the distribution. Hence, we can implement the following algorithm

- ▶ For $i = 1, \dots, m$: $z^{(i)} \sim P(z | \theta^{(n)})$
- ▶ $d^{(n)} = \frac{1}{m} \sum_{i=1}^m P(x | z^{(i)}, \theta) \nabla_{\theta} \ln P(z^{(i)}, x | \theta^{(n)})$
- ▶ $\theta^{(n+1)} = \theta^{(n)} + \alpha d^{(n)}$.

Gradients of the GMMs

For the Gaussian mixture model, we need to calculate the following gradient, with respect to $\theta = (\theta, \mu, \sigma)$.

$$\nabla_{\theta} \ln P(z, x | \theta) = \sum_{t=1}^T \nabla_{\theta} \ln P(z_t, x_t | \theta) = \sum_{t=1}^T \nabla_{\theta} \ln [P(z_t | \theta) P(x_t | z_t, \mu, \sigma)]$$

The individual terms are:

$$\nabla_{\theta} \ln P(z_t | \theta) = \frac{1}{P(z_t | \theta)} \nabla_{\theta} P(z_t | \theta) \quad \frac{\partial}{\partial \theta_i} = \frac{\mathbb{I}\{z_t = i\}}{P(z_t | \theta)}$$

$$\frac{\partial}{\partial \mu_i} \ln P(x_t | z_t = i, \mu, \sigma) = -\frac{\partial}{\partial \mu_i} \frac{(x_t - \mu_i)^2}{2\sigma_i^2} = \sigma_i^{-2} (x_t - \mu_i)$$

So the algorithm

- ▶ Randomly assigns z_t according to the current θ estimate
- ▶ Moves the centers towards the mean indicated by the z .

A lower bound on the likelihood

For any distribution $G(z)$, and specifically for $G(z) = P(z|x, \theta^{(k)})$:

$$\begin{aligned}
 \ln P(x|\theta) &= \sum_z G(z) \ln P(x|\theta) = \sum_z G(z) \ln[P(x, z|\theta)/P(z|x, \theta)] \\
 &= \sum_z G(z) [\ln P(x, z|\theta) - \ln P(z|x, \theta)] \\
 &= \sum_z G(z) \ln P(x, z|\theta) - \sum_z G(z) \ln P(z|x, \theta) \\
 &= \sum_z P(z|x, \theta^{(k)}) \ln P(x, z|\theta) - \sum_z P(z|x, \theta^{(k)}) \ln P(z|x, \theta) \\
 &\geq \sum_z P(z|x, \theta^{(k)}) \ln P(x, z|\theta) - \sum_z P(z|x, \theta^{(k)}) \ln P(z|x, \theta^{(k)}) \\
 &= Q(\theta | \theta^{(k)}) + \mathbb{H}(z | x, \theta^{(k)}),
 \end{aligned}$$

where

$$\mathbb{H}(z | x, \theta^{(k)}) = \sum_z P(z | x, \theta^{(k)}) \ln P(z | x, \theta^{(k)})$$

is the entropy of z for a fixed $x, \theta^{(k)}$. As this is not negative,
 $\ln P(x|\theta) \geq Q(\theta | \theta^{(k)})$.

Some information theory

Information theory notation can be a bit confusing. Sometimes we talk about random variables ω , and sometimes about probability measures P . This is context-dependent.

Entropy

For a random variable ω under distribution P , we denote the entropy as

$$\mathbb{H}_P(\omega) \equiv \mathbb{H}(P) \equiv \mathbb{H}(\omega) = \sum_{\omega \in \Omega} P(\omega) \ln P(\omega).$$

KL Divergence

For two probabilities P, Q over random outcomes in the same space Ω , we define

$$D_{KL}(P\|Q) = \sum_{\omega \in \Omega} P(\omega) \ln \frac{P(\omega)}{Q(\omega)}$$

The Gibbs Inequality

$$D_{KL}(P\|Q) \geq 0, \text{ or } \sum_x \ln P(x)P(x) \geq \sum_x \ln Q(x)P(x).$$

EM Algorithm (Dempster et al, 1977)

- ▶ Initial parameter $\theta^{(0)}$, observed data x
- ▶ For $k = 0, 1, \dots$
- Expectation step:

$$Q(\theta \mid \theta^{(k)}) \triangleq \mathbb{E}_{z \sim P(z|x, \theta^{(k)})} [\ln P(x, z | \theta)] = \sum_z [\ln P(x, z | \theta)] P(z \mid x, \theta^{(k)})$$

- Maximisation step:

$$\theta^{(k+1)} = \arg \max_{\theta} Q(\theta, \theta^{(k)}).$$

See *Expectation-Maximization as lower bound maximization*, Minka, 1998

Minorise-Maximise

EM can be seen as a version of the minorise-maximise algorithm

- ▶ $f(\theta)$: Target function to **maximise**
- ▶ $Q(\theta|\theta^{(k)})$: surrogate function

Q Minorizes f

This means surrogate is always a lower bound so that

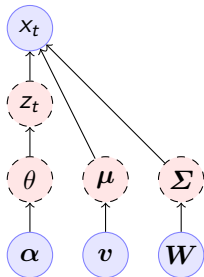
$$f(\theta) \geq Q(\theta|\theta^{(k)}), \quad f(\theta^{(k)}) = Q(\theta^{(k)}|\theta^{(k)}),$$

Algorithm

- ▶ Calculate: $Q(\theta|\theta^{(k)})$
- ▶ Optimise: $\theta^{(k+1)} = \arg \max_{\theta} Q(\theta|\theta^{(k)})$.

Bayesian inference for the GMM

We saw, for the Beta-Bernoulli and Dirichlet-Multinomial models, that a **conjugate prior** turns Bayesian inference into a simple parameter update. The same idea extends to the GMM, by placing a prior on every parameter.



$$\mu_i \mid \mathbf{v} \sim \text{Normal}(\mathbf{v}, I), \quad (5)$$

$$\Sigma_i \mid \mathbf{W} \sim \text{Wish}(\mathbf{W}), \quad (6)$$

$$\theta \mid \alpha \sim \text{Dir}(\alpha), \quad (7)$$

$$z_t \mid \theta \sim \text{Mult}(\theta), \quad (8)$$

$$x_t \mid \mu, \Sigma, z_t = i \sim \text{Normal}(\mu_i, \Sigma_i). \quad (9)$$

Exercise

Which of the five lines above, if any, would have a closed-form conjugate update if all the **others** were held fixed? Compare with the Beta-Bernoulli and Dirichlet-Multinomial cases covered earlier.

Graphical models

Random variables and probabilities

Graphical model

Conditional independence

Example Machine Learning Tasks

Classification: Discriminative modelling

Classification: Generative modelling

Density estimation

Algorithms for latent variable models

Gradient algorithms

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Classification

GMM versus histogram

- ▶ Generate some data x from an arbitrary distribution in $\mathbb{R}$.
- ▶ Fit the data with a histogram for varying numbers of bins
- ▶ Fit a GMM with varying numbers of Gaussians
- ▶ What is the best fit? How can you measure it?

GMM Classifier

Base class: sklearn GaussianMixtureModel

- ▶ *fit()* only works for Density Estimation
- ▶ *predict()* only predicts cluster labels

Problem

- ▶ Create a GMMClassifier class
- ▶ *fit()* should take X, y, arguments
- ▶ *predict()* should predict class labels
- ▶ Hint: Use *predict_proba()* and multiple GMM models